Лабораторна работа №2

$$\frac{dx}{dt} = 2tx + u \quad , \quad t \in [0, T]$$

$$x(0) = x_0 \qquad x(T) = y_0$$

1) $u(t) = C$, C - константа

$$\frac{dx}{dt} - 2tx = C \quad ; \quad \frac{dx}{dt} - 2tx = 0 \Rightarrow \frac{dx}{dt} = 2tx$$

$$\int \frac{dx}{x} = \int 2t\,dt$$

$$\ln|x| = 2\frac{t^2}{2} + \ln|C|$$

$$\ln|x| - \ln|C| = t^2 \Rightarrow \ln\left|\frac{x}{C}\right| = t^2$$

$$\left|\frac{x}{C}\right| = e^{t^2} \Rightarrow x_0 = Ce^{t^2}$$

$$x(t) = C(t)e^{t^2}$$

$$\frac{dx}{dt} = C'(t)e^{t^2} + 2tC(t)e^{t^2}$$

$$c'(t)e^{t^2} + \cancel{2tc(t)e^{t^2}} - \cancel{2tc(t)e^{t^2}} = c$$

$$c'(t)e^{t^2} = c$$

$$c'(t) = ce^{-t^2} \Rightarrow \int_0^t c'(s)\,ds = \int_0^t ce^{-s^2}\,ds$$

$$c(t) - c(0) = c\int_0^t e^{-s^2}\,ds$$

$$c(t) = c(0) + c\int_0^t e^{-s^2}\,ds$$

$$x(t) = \left(c(0) + c\int_0^t e^{-s^2}\,ds\right)e^{t^2}$$

$$t = 0 \Rightarrow x(0) = c(0)e^0 = c(0) \Rightarrow c(0) = x_0$$

$$\boxed{x(t) = e^{t^2}\left(x_0 + c\int_0^t e^{-s^2}\,ds\right)}$$

$$t = T \Rightarrow x(T) = e^{T^2}\left(x_0 + c\int_0^t e^{-s^2}\,ds\right) = y_0 \Rightarrow$$

$$\Rightarrow x_0 + c\int_0^t e^{-s^2}\,ds = y_0 e^{-T^2} \Rightarrow c\int_0^t e^{-s^2}\,ds = y_0 e^{-T^2} - x_0 \Rightarrow$$

$$\Rightarrow \boxed{c = \frac{y_0 e^{-T^2} - x_0}{\int_0^T e^{-s^2}\,ds}}$$

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2) $u(t) = ct$

$$\frac{dx}{dt} - 2tx = ct \Rightarrow x_0(t) = Ce^{t^2} \Rightarrow$$

$$\Rightarrow x(t) = C(t)e^{t^2}$$

$$\frac{dx}{dt} = C'(t)e^{t^2} + 2tC(t)e^{t^2} \Rightarrow C'(t)e^{t^2} + \cancel{2tC(t)e^{t^2}} - \cancel{2tC(t)e^{t^2}} = ct$$

$$\Rightarrow C'(t)e^{t^2} = ct \Rightarrow C'(t) = cte^{-t^2}$$

$$\int_0^t C'(s)ds = \int_0^t cse^{-s^2}ds \Rightarrow C(t) - C(0) = c\int_0^t se^{-s^2}ds \Rightarrow$$

$$\int_0^t se^{-s^2}ds = \left| \begin{array}{l} z = -s^2, \; dz = -2s\,ds \\ s\,ds = -\frac{1}{2}dz \end{array} \right| \Rightarrow$$

$$\Rightarrow \int_0^{-t^2} e^z\left(-\frac{1}{2}dz\right) = -\frac{1}{2}\cdot e^z\Big|_0^{-t^2} = -\frac{1}{2}\cdot\left(e^{-t^2} - e^0\right) =$$

$$= -\frac{1}{2}\left(e^{-t^2} - 1\right) = \frac{1-e^{-t^2}}{2}$$

(=>) $c(t) - c(0) = c\frac{1-e^{-t2}}{2} \Rightarrow c(t) = c(0) + \frac{c}{2}(1-e^{-t2})$

$$x(t) = \left(c(0) + \frac{c}{2}(1-e^{-t2})\right)e^{t2} \Rightarrow x(t) = c(0)e^{t2} + \frac{c}{2}(e^{t2}-1)$$

$t=0 \Rightarrow x(0) = c(0)e^0 + \frac{c}{2}(e^0 - 1) = c(0) - 1 + \frac{c}{2}(1-1) = c(0) \Rightarrow$

$\Rightarrow c(0) = x_0 \Rightarrow x(t) = x_0 e^{t2} + \frac{c}{2}(e^{t2} - 1)$

$t=T \Rightarrow x(T) = x_0 e^{T^2} + \frac{c}{2}(e^{T^2} - 1) = y_0$

$$\frac{c}{2}(e^{T^2} - 1) = y_0 - x_0 e^{T^2} \Rightarrow \boxed{C = \frac{2(y_0 - x_0 e^{T^2})}{e^{T^2} - 1}}$$

$$\boxed{x(t) = x_0 e^{t2} + \frac{y_0 - x_0 e^{T^2}}{e^{T^2} - 1}(e^{t2} - 1)}$$

3) $u(t) = cx$

$$\frac{dx}{dt} = 2tx + cx \Rightarrow \frac{dx}{dt} = (2t + c)x \Rightarrow \frac{dx}{x} = (2t + c)dt$$

$$\int \frac{dx}{x} = \int (2t + c)\,dt \Rightarrow \ln|x| = x\frac{t^2}{t} + ct + \ln|C| \Rightarrow$$

$$\Rightarrow \ln|x| - \ln|C| = t^2 + ct \Rightarrow \ln\left|\frac{x}{C}\right| = t^2 + ct \Rightarrow$$

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$$\Rightarrow \frac{x}{C} = e^{t^2+ct} \Rightarrow x(t) = Ce^{t^2+ct}$$

$$t=0 \Rightarrow x(0) = Ce^{0^2+c\cdot 0} = Ce^0 = C \Rightarrow x(0) = C$$

$$\underline{x(t) = x_0 e^{t^2+ct}}$$

$$t=T \Rightarrow x(T) = x_0 e^{T^2+cT} = y_0 \Rightarrow e^{T^2+cT} = \frac{y_0}{x_0}$$

$$\ln\left(e^{T^2+cT}\right) = \ln\left(\frac{y_0}{x_0}\right) \Rightarrow T^2+cT = \ln\left(\frac{y_0}{x_0}\right) \Rightarrow$$

$$\Rightarrow cT = \ln\left(\frac{y_0}{x_0}\right) - T^2 \Rightarrow \underline{c = \frac{1}{T}\ln\left(\frac{y_0}{x_0}\right) - T}$$

$$x(t) = x_0 e^{t^2 + \left(\frac{1}{T}\ln\left(\frac{y_0}{x_0}\right) - T\right)t}$$

$$u(t) = x\left(\frac{1}{t}\ln\left(\frac{y_0}{x_0}\right) - T\right)$$

ч) $$u(t) = \begin{cases} c_1, & t \in [0, t_1] \\ c_2, & t \in [t_1, T] \end{cases}$$

$$t \in [0, t_1], \; u(t) = c_1 \Rightarrow \frac{dx}{dt} - 2tx = c_1$$

$$x(t) = e^{t^2}\left(x_0 + c_1 \int_0^t e^{-s^2} ds\right) \Rightarrow x(t_1) = e^{t_1^2}\left(x_0 + c_1 \int_0^{t_1} e^{-s^2} ds\right)$$

$t \in [t_1, T]$

$$\frac{dx}{dt} - 2tx = c_2 \Rightarrow x(t) = c(t)e^{t^2} \Rightarrow c'(t)e^{t^2} = c_2 \Rightarrow$$

$$\Rightarrow c'(t) = c_2 e^{-t^2}$$

$$\int_{t_1}^{t} c'(s)ds = \int_{t_1}^{t} c_2 e^{-s^2} ds \Rightarrow c(t) - c(t_1) = c_2 \int_{t_1}^{t} e^{-s^2} ds \Rightarrow$$

$$\Rightarrow c(t) = c(t_1) + c_2 \int_{t_1}^{t} e^{-s^2} ds$$

$$c(t_1) = x(t_1) e^{-t_1^2}$$

$$c(t_1) = e^{t_1^2} \left(x_0 + c_1 \int_0^{t_1} e^{-s^2} ds \right) e^{-t_1^2} = x_0 + c_1 \int_0^{t_1} e^{-s^2} ds$$

$$c(t) = x_0 + c_1 \int_0^{t_1} e^{-s^2} ds + c_2 \int_{t_1}^{t} e^{-s^2} ds$$

$$x(t) = e^{t^2} \left(x_0 + c_1 \int_0^{t_1} e^{-s^2} ds + c_2 \int_{t_1}^{t} e^{-s^2} ds \right)$$

$$x(T) = x_0.$$

$$x(T) = e^{T^2} \left(x_0 + c_1 \int_0^{t_1} e^{-s^2} ds + c_2 \int_{t_1}^{T} e^{-s^2} ds \right) = x_0$$

$$C_1\int_0^{t_1} e^{-s^2}ds + C_2\int_{t_1}^{T} e^{-s^2}ds = y_0 e^{-T^2} - x_0$$

$$I_1 = \int_0^{t_1} e^{-s^2}ds > 0$$

$$I_2 = \int_{t_1}^{T} e^{-s^2}ds > 0 \qquad \Rightarrow C_1 I_1 + C_2 I_2 = y_0 e^{-T^2} - x_0$$

$$C_1 \in \mathbb{R}\ ,\ C_2 = \frac{y_0 e^{-T^2} - x_0 - C_1 I_1}{I_2}$$

$$x(t) = \begin{cases} e^{t^2}\left(x_0 + C_1\int_0^t e^{-s^2}ds\right)\ , & t\in[0,t_1] \\ e^{t^2}\left(x_0 + C_1\int_0^{t_1} e^{-s^2}ds + C_2\int_{t_1}^t e^{-s^2}ds\right)\ , & t\in[t_1,T] \end{cases}$$